LEARNING MODULE

31

Risk Parity, HRP and Volatility Targeting

When means cannot be trusted, allocate by risk alone: equal risk contributions, a tree of correlations, and a book scaled by a volatility forecast.

VOLUME V · Portfolio Optimization

BUILDS ON LM7 · LM8 · LM10 · LM11 · LM16 · LM26 · LM27 · LM28 · LM29 · LM30

LEADS TO LM32 · LM33 · LM34 · LM35 **TIME** ≈ 3 h + lab

LEARNING OUTCOMES

Mastery	*After this module you should be able to:*
☐	1 derive the budget conditions (31.1) and show when ERC gives inverse-volatility weights
☐	2 prove that a long-only risk-budget portfolio exists and is unique when Σ is positive definite
☐	3 compute equal-risk and risk-budget weights by Newton's method and coordinate descent, and check them with cvxpy
☐	4 build an HRP portfolio from a correlation matrix, and explain why implementations can give different trees
☐	5 explain what risk-based weights assume, and why risk shares set in advance differ after the fact
☐	6 compute a volatility-targeted exposure from an EWMA forecast, with its cap, cash and borrowing
☐	7 judge a volatility target out of sample, and explain how periodic and risk-triggered rebalancing differ
☐	8 state the evidence on volatility-managed portfolios with its sample and design

LEARNING MODULE OVERVIEW

- With two funds, equal risk contributions (ERC) give inverse-volatility weights whatever the correlation. With five they do not: on the first window inverse volatility leaves the bond fund 00679B only 9.8% of the risk.
- A long-only risk-budget portfolio exists and is unique for a positive definite covariance matrix and positive budgets: it is the normalized minimizer of a strictly convex function. Newton's method finds R6's first weights in 6 steps and coordinate descent in 12 sweeps; a fixed point with equal shares settles at negative weights.
- HRP links funds by single linkage on the distance between columns of $\sqrt{(1-\rho_{ij})/2}$. Linking on that distance itself changes the quasi-diagonal order at 17 of 92 signals.
- On U28's second half, with weights from the first, HRP's volatility, 14.96%, is the lowest of 1/N and the three risk-based portfolios, and above both minimum-variance portfolios.
- Out of sample R6 and R7 had volatilities of 10.32% and 10.14%. The mean of R6's target weights carried 18.0% to 23.1% of the realized risk, not 20% each.
- R8 scales R6 by $\min(1.5,\ \bar{\sigma}_t/\sigma_{t+1|t})$: its exposure averaged 1.14 over 92 trades and was at the cap at 14 of them.
- Over 91 holding months the median ratio of realized to target volatility rose from R6's 0.83 to R8's 0.98, while the SD of its logarithm stayed at 0.421. R8 entered April 2025 at 1.33 and realized 5.78 times its target.
- The EWMA forecast beat the window's volatility in 56 of 91 holding months, yet its mean QLIKE was the higher, 0.791 against 0.691, because of one holding month.

1 INTRODUCTION

LM27 split a book's volatility into risk contributions. LM28 to LM30 traded mean against variance, and their minimum-variance portfolios used the covariance matrix alone. This module uses risk alone in three new ways. It gives each fund a chosen share of the risk, builds a portfolio from a tree of correlations, and scales the whole book by a volatility forecast.

MOTIVATING CASE **Equal risk, then a forecast**

An investor who cannot trust estimated means can still share risk evenly. On 31 March 2025 rule R6 did that for E5's five funds (LM27). It put 30.7% in the bond fund 00679B and 12.0% in the oil-futures fund 00642U, so each fund carried a fifth of the risk. Rule R8 then scales R6 by a forecast.

R6's book had been calm: its EWMA forecast was 5.89% a year, against 7.84% over the 504 joint dates before it. So on 1 April R8 held 1.33 times its equity in the funds, borrowing the extra 0.33 at 3.5% a year. On 7 April, the first session after the Qingming holiday, the forecast jumped to 23.58%, an implied exposure of 0.33. R8 trades only after a month-end, so it kept its leverage until 2 May.

Over that holding month, 2 April to 2 May (Section 7), its realized volatility was 5.78 times its target (Exhibit 1). That ratio was the worst of all 91. Does a forecast keep risk steady? And what does an equal share of risk buy?

Exhibit 1: R6 and R8 around April 2025

Date	Quantity	Value
31 March 2025, signal	R6's weights: 0050, 00646, 00679B, 00635U, 00642U	13.5, 21.6, 30.7, 22.2, 12.0%
	window volatility $\bar{\sigma}_t$ of R6's book	7.84%
	EWMA forecast $\sigma_{t+1\|t}$	5.89%
1 April, trade	R8's exposure L_t	1.33
7 April	EWMA forecast, continued daily on R6's 31 March weights	23.58%
	the exposure it implies, $\bar{\sigma}_t$ over that forecast	0.33
2 May, next trade	R8's exposure	0.45
2 April → 2 May	R8's realized volatility	45.3%
	its ratio to the target, the largest of 91	5.78
91 holding months	median ratio, R6 and R8	0.83, 0.98

Note: out-of-sample, fixed in advance on 24 October 2026; volatilities annualized with $A = 242.95$, so the ratio is the printed figures'. EWMA (26.2), $\lambda = 0.94$, on the daily returns of R6's 31 March weights held constant. Source: FinMind `TaiwanStockPrice`, `TaiwanStockDividendResult` and `TaiwanStockSplitPrice`, retrieved 24 October 2026. Computations: `data/lm31_numbers.py` (through `data/vol5_rules.py`).

Sections 2 to 5 treat R6 and R7, Sections 6 and 7 treat R8, and Section 8 the code.

NOTE **Notation for this module**

E5 is LM27's: $T = 2{,}362$ joint one-session returns from 18 January 2017 to 8 October 2026, $A = 242.95$; the wealth frame has 2,368 sessions, $A_{\mathrm{w}} = 243.57$, over 9.7221 years. U28 is LM10's 28 sector price indices, $T = 4{,}382$, $A = 245.48$. Covariances divide by T; λ is LM26's EWMA decay only.

Local: $\bar{\rho}$ a mean correlation; $\mathbf{b}$ risk budgets; $\mathbf{y}$ positions, $\mathbf{w} = \mathbf{y}/\mathbf{1}^{\top}\mathbf{y}$; h of (31.2); σ_{mv}, σ_{erc}, $\sigma_{1/N}$ the volatilities of long-only minimum variance, ERC and 1/N ($\sigma_{\min}$ of (28.2) when $\mathbf{w}^{\mathrm{mv}} \geq \mathbf{0}$); d_{ij} HRP's distance, $\mathbf{d}_j$ column j of (d_{ij}); $\sigma^2_{(1)}$, $\sigma^2_{(2)}$ a bisection's variances; $\bar{\sigma}_t$, $\sigma_{t+1|t}$ R8's window volatility and forecast; $\Sigma_{t+1|t}$, Σ_{start} the EWMA matrix and its start; $\mathbf{w}_t^{\mathrm{R6}}$ R6's weights; V a holding month's mean squared return. Counts: i, j funds, t dates, n list items.

NOTE **Research hygiene**

R6, R7 and R8 are three of the eight rules fixed on 24 October 2026, before any result was computed. Their weights use only the 504 joint dates up to and including each month-end. They trade at the next close on which every fund trades (LM27). So every through-case result is **out-of-sample**. Whether one rule beats another is LM33's test, across eight rules.

Weights, risk shares and trees fitted to one window are **in-sample** plug-in estimates, with sampling error not measured here. U28's halves, one **out-of-sample** split fixed in advance, compare three risk-based portfolios with LM30's 1/N and two minimum-variance portfolios. Six other versions are **in-sample** illustrations, not candidates. They are inverse volatility, R2's mix as risk budgets, the tree linked on d, the fixed point, and the lab's triggered and GARCH-scaled R8. The widget's and the problems' inputs are **constructed**.

2 RISK CONTRIBUTIONS AS TARGETS

- [ ] derive the budget conditions (31.1) and show when ERC gives inverse-volatility weights

- [ ] explain what risk-based weights assume, and why risk shares set in advance differ after the fact

On the first window, 13 February 2017 to 27 February 2019, R1 held 20% of each fund. Yet 00642U carried 50.2% of its risk (Exhibit 2), as LM27's note on its trailing windows also shows. For $\sigma_p > 0$, (27.5) and (27.6) give

$$\mathrm{MRC}_i = \frac{(\Sigma \mathbf{w})_i}{\sigma_p}, \qquad \mathrm{RC}_i = w_i \, \mathrm{MRC}_i,$$
$$\sum_i \mathrm{RC}_i = \sigma_p.$$

Here MRC_i is LM7's marginal variance divided by $2\sigma_p$, and RC_i/σ_p is fund i's **risk share**.

DEFINITION **Risk budgets**

Let Σ be positive definite, and let budgets $b_i > 0$ sum to 1. A fully invested portfolio, $\mathbf{1}^\top \mathbf{w} = 1$, meets them when $\mathrm{RC}_i = b_i \sigma_p$ for every i. Multiplying by σ_p and using (27.6), that is

$$w_i (\Sigma \mathbf{w})_i = b_i \, \mathbf{w}^\top \Sigma \mathbf{w}, \qquad i = 1, \dots, N. \tag{31.1}$$

The **risk-budget portfolio** is the long-only solution, $\mathbf{w} \geq \mathbf{0}$, which exists and is unique by (31.2), Section 3. **Equal risk contributions** (ERC) take $b_i = 1/N$.

EXAMPLE 1

Two funds

On the first window 0050 and the bond fund 00679B have volatilities 13.50% and 8.54% and correlation −0.30 (LM28). Find their ERC weights.

SOLUTION

With two funds, (31.1) at $b_i = \tfrac{1}{2}$ reads

$$w_1 (w_1 \sigma_1^2 + w_2 \sigma_{12}) = w_2 (w_1 \sigma_{12} + w_2 \sigma_2^2).$$

The covariance terms cancel, so $w_1^2 \sigma_1^2 = w_2^2 \sigma_2^2$, and positive weights give $w_1 \sigma_1 = w_2 \sigma_2$ whatever ρ. So $w_1 = 8.54/(13.50 + 8.54) = 0.3875$ and $w_2 = 0.6125$, with risk shares 50/50. LM7's minimum-variance weight (7.7) puts 0.331 in 0050, at 6.1% against ERC's 6.2%.

RESULT **When ERC is inverse volatility (Maillard, Roncalli and Teïletche 2010)**

Let Σ be positive definite. The inverse-volatility weights

$$w_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j} \tag{31.3}$$

are an ERC portfolio for two assets, whatever their correlation, and for N assets whose correlations all equal ρ, with $-1/(N-1) < \rho < 1$. By (31.2), Section 3, they are then the only long-only one.

Proof. Two assets are Example 1; at $\rho = -1$ the book's σ_p would be 0. With one correlation ρ and the weights (31.3),

$$(\Sigma\mathbf{w})_i = \sum_j \rho_{ij}\sigma_i\sigma_j w_j = \frac{\sigma_i\big(1+(N-1)\rho\big)}{\sum_j 1/\sigma_j},$$

so $w_i(\Sigma\mathbf{w})_i$ is the same for every i. The correlation matrix has eigenvalues $1+(N-1)\rho$ and $1-\rho$ (LM11), so Σ is positive definite exactly when $-1/(N-1) < \rho < 1$. ■

By (29.5), each fund that long-only minimum variance holds has the rate $(\Sigma\mathbf{w})_i = \sigma_p^2$, so its beta $\beta_{i,p}$ is 1 and its share $w_i\beta_{i,p}$ is its weight. ERC equalizes the products $w_i(\Sigma\mathbf{w})_i$ instead. The rate $(\Sigma\mathbf{w})_i$ is half of LM10's gradient entry, and a share is a weight times that rate over σ_p^2. Maillard, Roncalli and Teïletche (2010) order the volatilities:

$$\sigma_{\text{mv}} \le \sigma_{\text{erc}} \le \sigma_{1/N},$$

with σ_{mv} the volatility of long-only minimum variance (stated without proof; their 2009 working paper, Appendix A.3). Bruder and Roncalli (2012, §2.3.4, p. 10) put a risk-budget portfolio between minimum variance and the budgets used as weights (stated without proof; their Appendix A.3).

EXAMPLE 2

Five funds on the first window

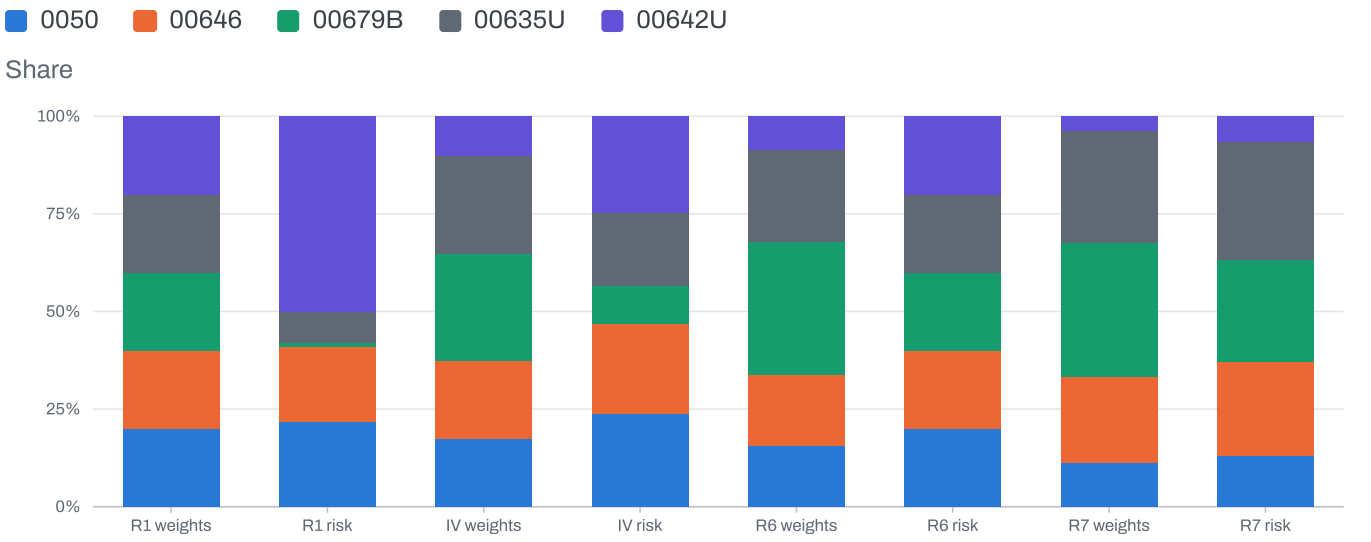
The five funds' volatilities are 13.50, 11.87, 8.54, 9.39 and 23.17% (0050, 00646, 00679B, 00635U, 00642U; LM28). Compare inverse volatility with ERC.

SOLUTION

Inverse-volatility weights are 0.174, 0.198, 0.275, 0.251 and 0.102. With them every $w_i\sigma_i$ is equal, so each share is proportional to the fund's row sum of correlations, as in the proof of (31.3). 00679B correlates −0.30 and −0.26 with the two equity funds, so its row sum is the lowest and its share is 9.8%. 00642U's row sum, the highest, gives 24.8%. R6 holds 0.1561, 0.1828, 0.3386, 0.2372 and 0.0854 (rounded separately; 1.0001 as printed), and each fund carries 20%.

The volatilities, 5.02% for R3, 5.45% for R6 and 7.38% for R1 (R3's and R1's from LM28), follow Maillard et al.'s order. R3 equals (28.2)'s $\mathbf{w}^{\text{mv}}$ here, since every weight of $\mathbf{w}^{\text{mv}}$ is positive. With every correlation set to the mean, $\bar\rho = 0.0533$, ERC equals inverse volatility. All use the window's unrounded $\hat\Sigma$.

Exhibit 2: Weights and risk shares on the first window



Note: first window; in-sample. IV is inverse volatility; 00635U is grey. R1's 00642U share is 50.2% (LM27). IV's shares are 23.8, 23.0, 9.8, 18.7 and 24.8%, and R7's 13.0, 24.0, 26.3, 30.0 and 6.6%; rounded separately, they sum to 100.1 and 99.9. Source: as Exhibit 1.

PITFALL **A risk share is neither a weight nor a share of losses**

R6 holds 33.9% in 00679B, which carries 20% of its risk. A risk share splits one covariance matrix's variance before the fact. It is no share of a day's loss, and it moves afterwards (Section 5).

導讀 **風險預算：每檔基金分擔指定的風險**

權重相等不代表風險相等：第一個視窗裡，等權重投資組合有一半的風險來自石油期貨基金。風險貢獻（risk contribution）把投資組合波動率拆成各資產的部分，加總剛好是投資組合波動率。風險預算（risk budget）事先指定每檔資產分擔的比例，每檔都是 $1/N$ 時稱為等風險貢獻（equal risk contribution，ERC），是最常見的風險平價投資組合（risk-parity portfolio）。本模組的風險預算是分給每一檔資產的份額，LM7 的風險預算則是兩個策略合計的波動率上限，兩者不同。最小變異數投資組合（minimum-variance portfolio）讓持有資產的 $(\Sigma \mathbf{w})_i$ 相等，ERC 讓權重乘上它的乘積相等，所以低波動又與股票負相關的債券基金權重很大，風險份額（risk share）卻只有五分之一。

KNOWLEDGE CHECK 1

On the first window R3 holds 41.4% in 00679B, and 41.4% of its risk is there (LM28). R6 holds 33.9% in 00679B, yet that fund carries only 20% of R6's risk. Why less than its weight?

ANSWER

Minimum variance equalizes the rates $(\Sigma \mathbf{w})_i$ of held funds, (29.5), so each share equals its weight; ERC equalizes the products $w_i(\Sigma \mathbf{w})_i$. 00679B's rate is low, since it is the calmest fund (8.5% a year, LM28) and moves against the equity funds. So a large weight buys a fifth of the risk.

3 BUDGETS: EXISTENCE, UNIQUENESS AND COMPUTATION

- ☐ prove that a long-only risk-budget portfolio exists and is unique when Σ is positive definite
- ☐ compute equal-risk and risk-budget weights by Newton's method and coordinate descent, and check them with cvxpy

A change of variables makes the N nonlinear equations (31.1) the first-order conditions of a convex problem.

THEOREM The risk-budget portfolio (Bruder and Roncalli 2012, §2.3.3, pp. 9–10)

Let h be the function on the open set $\mathbf{y} > \mathbf{0}$ given by

$$h(\mathbf{y}) = \tfrac{1}{2}\mathbf{y}^\top \Sigma \mathbf{y} - \sum_i b_i \ln y_i,$$

and suppose that

1. the matrix Σ is positive definite;
2. every budget is positive, $b_i > 0$;
3. the budgets sum to 1.

Then h has exactly one minimizer $\mathbf{y}^*$, and its normalized form

$$\mathbf{w}^* = \frac{\mathbf{y}^*}{\mathbf{1}^\top \mathbf{y}^*} \tag{31.2}$$

is the only long-only solution of (31.1).

Proof. The Hessian of h is $\Sigma + \operatorname{diag}(b_i/y_i^2)$, which is positive definite on $\mathbf{y} > \mathbf{0}$. So h is strictly convex there by (29.2) and has at most one minimizer by (29.3)'s argument; one exists (Deep Dive). At it the gradient $\Sigma\mathbf{y} - \mathbf{b}/\mathbf{y}$ vanishes ($\mathbf{b}/\mathbf{y}$ entry by entry), so

$$y_i(\Sigma\mathbf{y})_i = b_i, \qquad \mathbf{y}^\top\Sigma\mathbf{y} = \sum_i b_i = 1.$$

Dividing both by $(\mathbf{1}^\top\mathbf{y}^*)^2$ shows that $\mathbf{w}^*$ satisfies (31.1). Conversely, let a long-only $\mathbf{w}$ with $\mathbf{1}^\top\mathbf{w} = 1$ solve (31.1). Each $w_i(\Sigma\mathbf{w})_i = b_i\mathbf{w}^\top\Sigma\mathbf{w}$ is then positive, so every $w_i > 0$, and $\mathbf{y} = \mathbf{w}/\sqrt{\mathbf{w}^\top\Sigma\mathbf{w}}$ has $y_i(\Sigma\mathbf{y})_i = b_i$. So the gradient of h vanishes there, and a stationary point of a convex function is its minimizer, by (29.2)'s tangent inequality. So $\mathbf{y} = \mathbf{y}^*$ and $\mathbf{w} = \mathbf{w}^*$; Bruder and Roncalli only sketch this argument. ■

> **DEEP DIVE** **Deep Dive: why the minimizer exists** optional · click to expand
>
> By (11.4), $\mathbf{y}^\top\Sigma\mathbf{y}$ is at least $\|\mathbf{y}\|^2$ times the smallest eigenvalue of Σ, which is positive. Each y_i is at most $\|\mathbf{y}\|$ and the budgets sum to 1, so $-\sum_i b_i \ln y_i \geq -\ln\|\mathbf{y}\|$. Hence $h \to \infty$ as $\|\mathbf{y}\| \to \infty$, and also as any $y_i \to 0^+$ on a bounded set. So the sublevel set $\{\mathbf{y} > \mathbf{0} : h(\mathbf{y}) \leq h(\mathbf{1})\}$ is closed and bounded, and the continuous h attains a minimum on it (LM11's extreme value theorem).

Newton's method, LM7's in five variables, solves a linear system for each step $\boldsymbol{\Delta}$:

$$\left(\Sigma + \operatorname{diag}(b_i/y_i^2)\right)\boldsymbol{\Delta} = -(\Sigma\mathbf{y} - \mathbf{b}/\mathbf{y}).$$

The rules' `erc()` starts at $y_i = 1/\sigma_i$ rescaled to $\mathbf{y}^\top\Sigma\mathbf{y} = 1/N$. It halves a step until $\mathbf{y} > \mathbf{0}$ and, while the largest gap $|y_i(\Sigma\mathbf{y})_i - b_i|$ exceeds 10^{-6}, until h falls enough. It stops when that gap is below 10^{-13}.

Cyclical coordinate descent (CCD; Griveau-Billion, Richard and Roncalli 2013) minimizes h along one y_i at a time. With the others fixed, $y_i\,\partial h/\partial y_i = 0$ is a quadratic whose positive root is the update:

$$\begin{aligned} \sigma_i^2 y_i^2 + \Big(\sum_{j\neq i}\sigma_{ij}y_j\Big)\,y_i - b_i = 0, \\ y_i \leftarrow \frac{\sqrt{\big(\sum_{j\neq i}\sigma_{ij}y_j\big)^2 + 4\sigma_i^2 b_i} - \sum_{j\neq i}\sigma_{ij}y_j}{2\sigma_i^2}. \end{aligned} \tag{31.4}$$

The sum is $(\Sigma\mathbf{y})_i - \sigma_i^2 y_i$. A sweep updates y_1 to y_N in turn, each update lowers h, and the sweeps converge (Tseng 2001, Theorem 5.1, as Griveau-Billion et al. cite it; stated without proof).

A fixed point. At the solution $w_i\beta_{i,p} = b_i$ by (27.6), so $w_i \propto b_i/\beta_{i,p}$ (Bruder and Roncalli, eq. (7), p. 9). Iterating that property gives the fixed-point iteration that Griveau-Billion et al. attribute to Chaves et al. (2012); it is no descent of h. cvxpy minimizes h as an exponential-cone program, a cone program of the kind LM29 names but does not develop. CLARABEL solves that program to about 10^{-6}, so it checks Newton to 10^{-5}.

EXAMPLE 3

R6's first weights four ways

Compute R6's weights on the first window's annualized $\hat{\Sigma}$ by Newton's method, CCD, cvxpy and the fixed point.

SOLUTION

Newton from `erc()` 's start takes 6 steps. CCD from $y_i = 1/\sigma_i$ takes 12 sweeps to $\max_i |y_i(\Sigma \mathbf{y})_i - \frac{1}{5}| < 10^{-12}$, within 3.5×10^{-13} of Newton. cvxpy, with CLARABEL's `tol_gap_abs` , `tol_gap_rel` and `tol_feas` at 10^{-10}, stays within 1.2×10^{-6} of Newton over the 92 windows. Daily or annualized, $\hat{\Sigma}$ gives the same weights to rounding.

The fixed point from 1/N, stopped when no weight moves by 10^{-12}, settles after 220 steps at −1.307, −1.306, 2.132, 2.408 and −0.927: every share 20%, three weights negative. At 45 of the 92 windows it does not settle in 10,000 steps.

EXAMPLE 4

R2's mix as risk budgets

R2 holds 40% 0050, 20% 00646, 30% 00679B, 10% 00635U and no 00642U. Use these as risk budgets for the four funds on the first window.

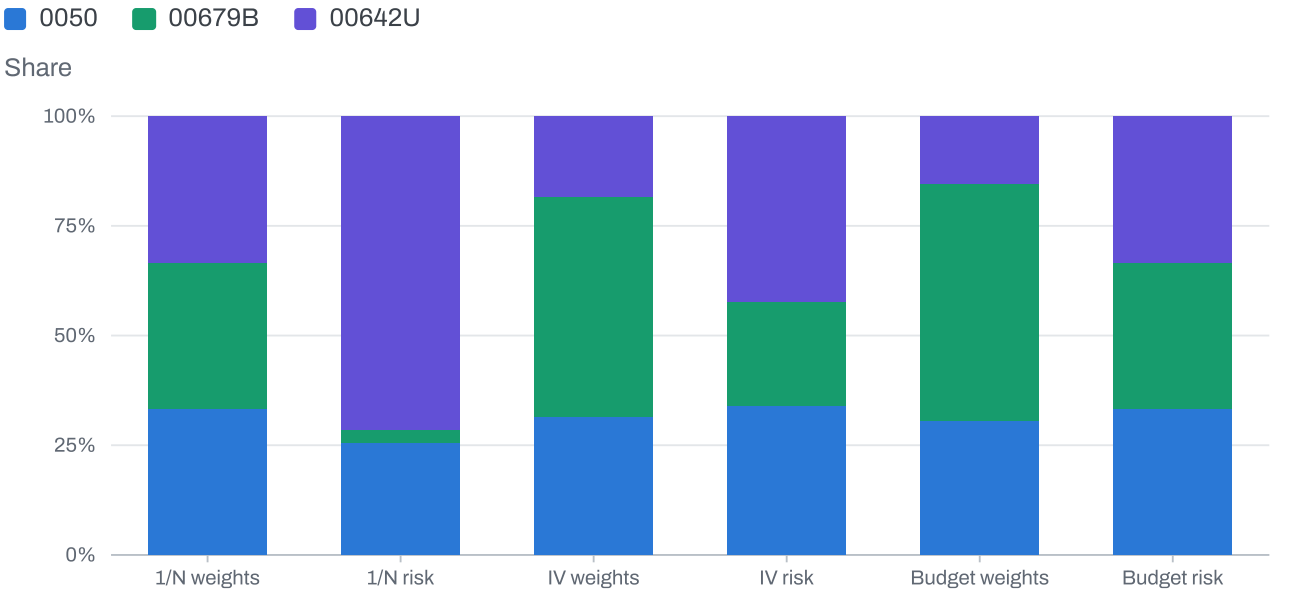
SOLUTION

A zero weight cannot be a budget, so 00642U is dropped. CCD gives 0.2466, 0.1773, 0.4190 and 0.1571 in 12 sweeps, and cvxpy agrees within 10^{-6}. So a 40% risk budget for 0050 is a 24.7% weight, while R2 puts 71.1% of its risk there. The volatilities follow Bruder and Roncalli's order: 5.016% for the four funds' minimum variance (above R3's 5.015%, from LM28's $\sigma^2_{\min} = 0.0025152$), 5.326% for the budgets and 6.709% for R2 (LM30).

The widget runs CCD (31.4) for three funds, from the first window's 0050, 00679B and 00642U with equal budgets (LM28), on unrounded inputs. There the fixed point settles at −32.5, 54.5 and −20.9 (rounded separately; 1.1 as printed).

Equal risk, or a budget, for three funds

INTERACTIVE



Budget weights	Budget risk shares	Inverse volatility's risk shares
0.3061, 0.5400, 0.1539	**33.3%, 33.3%, 33.3%**	**34.1%, 23.6%, 42.3%**
1/N's risk shares	Volatility a year: 1/N, IV, budget	CCD sweeps
25.7%, 2.8%, 71.5%	**9.51%, 6.93%, 6.55%**	**9**

Starting point (in-sample): the first window's 0050, 00679B and 00642U (13 February 2017 to 27 February 2019), with their unrounded volatilities and correlations and $b_1 = 1/3$, so the risk-budget portfolio is ERC. Once a slider moves, every input is a slider value (constructed). The other two funds share $1 - b_1$ equally. Weights by CCD (31.4) to 1e-12; risk shares RC_i/σ_p; volatilities a year.

PITFALL **A fixed point is not an algorithm**

The property $w_i \propto b_i/\beta_{i,p}$ holds at the solution, but iterating it need not settle, and it can settle at negative weights. Newton's method and CCD descend a strictly convex h. Check an answer by its risk shares and signs, not by a solver's status.

導讀 **先保證唯一，再談演算法**

風險預算條件 (31.1) 是一組非線性方程式，本身不保證只有一個解。把問題改寫成在 $\mathbf{y} > \mathbf{0}$ 上最小化 $h(\mathbf{y})$ 後，嚴格凸函數（strictly convex function）h 的最小值點最多只有一個；任一 y_i 趨近 0 或 $\|\mathbf{y}\|$ 趨近無窮大時 h 都趨近無窮大，所以最小值點一定存在（見 Deep Dive）。把這個最小值點正規化，就得到只作多（long-only）的風險預算投資組合（risk-budget portfolio），而且只作多的解只有這一個，也就是唯一性（uniqueness）。牛頓法（Newton's method）與座標下降法（coordinate descent）每一步都讓 h 下降；座標下降法的收斂引用 Tseng（2001）的定理，未加證明。不動點（fixed point）迭代 $w_i \propto b_i/\beta_{i,p}$ 不是下降法，可能停在含負權重的解，也可能一直不收斂。檢查答案要看風險份額（risk share）是否等於預算、權重是否全為正，不能只看 solver 回報的狀態。

KNOWLEDGE CHECK 2

R2 holds no 00642U. Can its weights serve as risk budgets for all five funds?

ANSWER

No. A zero budget removes that fund's log term from h, so (31.1) can have several solutions (Bruder and Roncalli 2012, §2.2, pp. 4–5), and theorem (31.2) does not apply. CCD's update (31.4) still runs, but it sets that y_i to 0 whenever $(\Sigma\mathbf{y})_i - \sigma_i^2 y_i \geq 0$, outside $\mathbf{y} > \mathbf{0}$. Drop the fund instead (Example 4).

4 HIERARCHICAL RISK PARITY

- [ ] build an HRP portfolio from a correlation matrix, and explain why implementations can give different trees

Hierarchical risk parity (HRP; López de Prado 2016) needs no optimizer and no inverse of Σ. It builds a tree of the assets from their correlations, orders the assets along it, and splits the weight from the top down. Its distance is half of LM8's per-date distance in (8.10):

$$d_{ij} = \sqrt{\tfrac{1}{2}(1-\rho_{ij})} = \tfrac{1}{2}\sqrt{2(1-\rho_{ij})}, \qquad \text{column distance } \|\mathbf{d}_i - \mathbf{d}_j\|. \tag{31.5}$$

LM8's distance is Euclidean, so d_{ij} satisfies the triangle inequality (8.8). HRP clusters on the **column distance**, the norm (8.5) of the difference between columns i and j of (d_{ij}). Assets are close when their distances to all assets are alike.

1. **Tree.** Single linkage merges the two closest clusters until one is left, the distance between clusters being the smallest distance between their members. Numbered from 1 in SciPy's order, the assets are 1 to N and the clusters $N+1$, $N+2$, ... as they form; each merge lists its lower-numbered member first.
2. **Order.** From the root down, replace each cluster by its two members, in that order. This quasi-diagonal order puts correlated assets side by side.
3. **Bisection.** Start with weight 1 on the ordered list. Split every list of $n > 1$ items into its first $\lfloor n/2 \rfloor$ items and the rest, until single assets remain. Each half's inverse-variance portfolio, weights in proportion to $1/\sigma_i^2$, has variance $\sigma_{(1)}^2$ or $\sigma_{(2)}^2$.

Each split multiplies the weights of the two halves by

$$\frac{\sigma_{(2)}^2}{\sigma_{(1)}^2 + \sigma_{(2)}^2} \quad \text{and} \quad \frac{\sigma_{(1)}^2}{\sigma_{(1)}^2 + \sigma_{(2)}^2}. \tag{31.6}$$

The calmer half gets more, and inverse variance is minimum variance for a diagonal covariance matrix (López de Prado, Appendix A.2). Single linkage only compares distances, so the factor 2 to LM8's distance doubles the heights and changes no tree. Other linkages, and splits along the tree's clusters (an option López de Prado notes), are not in this book.

EXAMPLE 5

HRP for three funds by hand

Build HRP for 0050, 00646 and 00679B on the first window, whose correlations are 0.54, −0.30 and −0.26 (LM28).

SOLUTION

By (31.5), d is 0.480, 0.806 and 0.794, and three decimals are kept from here on. The column distances are 0.679 (0050, 00646), 1.182 (0050, 00679B) and 1.169 (00646, 00679B). So 0050 and 00646 merge at 0.679, 00679B joins at $\min(1.182, 1.169) = 1.169$, and the order is (00679B, 0050, 00646).

The first cut is [00679B] | [0050, 00646], whose inverse-variance portfolios have variances 72.96 and 121.95 (%² a year), so 00679B takes $121.95/(72.96 + 121.95) = 0.6257$. In the second cut 0050's 182.29 meets 00646's 140.88, so 0050 takes 0.4359 of the remaining 0.3743. The weights are 0.1632, 0.2111 and 0.6257. The variances use the window's unrounded $\hat{\Sigma}$: squaring Example 1's 13.50% and 8.54% would give 182.25 and 72.93.

EXAMPLE 6

R7's first weights

Follow HRP on all five funds of the first window.

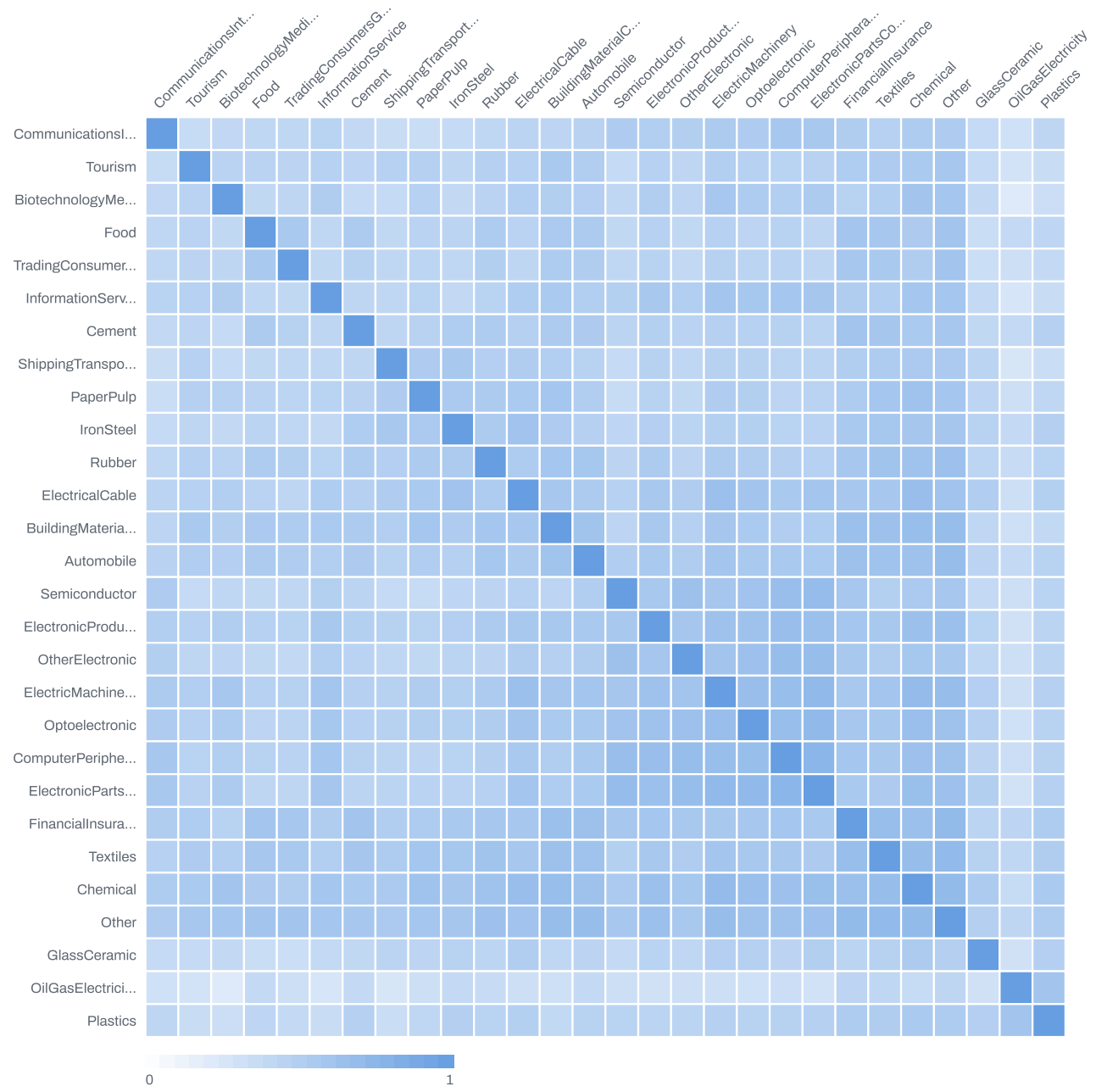
SOLUTION

The merges are 0050 with 00646 at 0.6835, then 00642U at 0.8576, 00679B with 00635U at 0.8861, and the root at 0.9822. The order is (00642U, 0050, 00646, 00679B, 00635U). The first cut, [00642U, 0050] | [00646, 00679B, 00635U], has a left share of 0.1516 and separates 0050 from 00646, the tree's first pair.

R7's weights are 0.1131, 0.2189, 0.3443, 0.2852 and 0.0384 (rounded separately; 0.9999 as printed, and 0.1515 for the left half), and its volatility is 5.13%. Its risk shares are 13.0, 24.0, 26.3, 30.0 and 6.6% (rounded separately; 99.9 as printed).

On U28 the 378 correlations run from 0.217 to 0.755 (Exhibit 3). The first merge joins ComputerPeripheralEquipment and ElectronicPartsComponents at 0.5012, and the root merges at 0.7426 (Exhibit 4). HRP's weights run from 12.87% (CommunicationsInternet) to 1.29% (GlassCeramic), ERC's from 6.06% to 2.43%. In sample HRP's volatility is 14.48% a year, against 15.18% for ERC, 15.28% for inverse volatility and 15.75% for 1/N (LM10).

Exhibit 3: U28's correlations in HRP's order

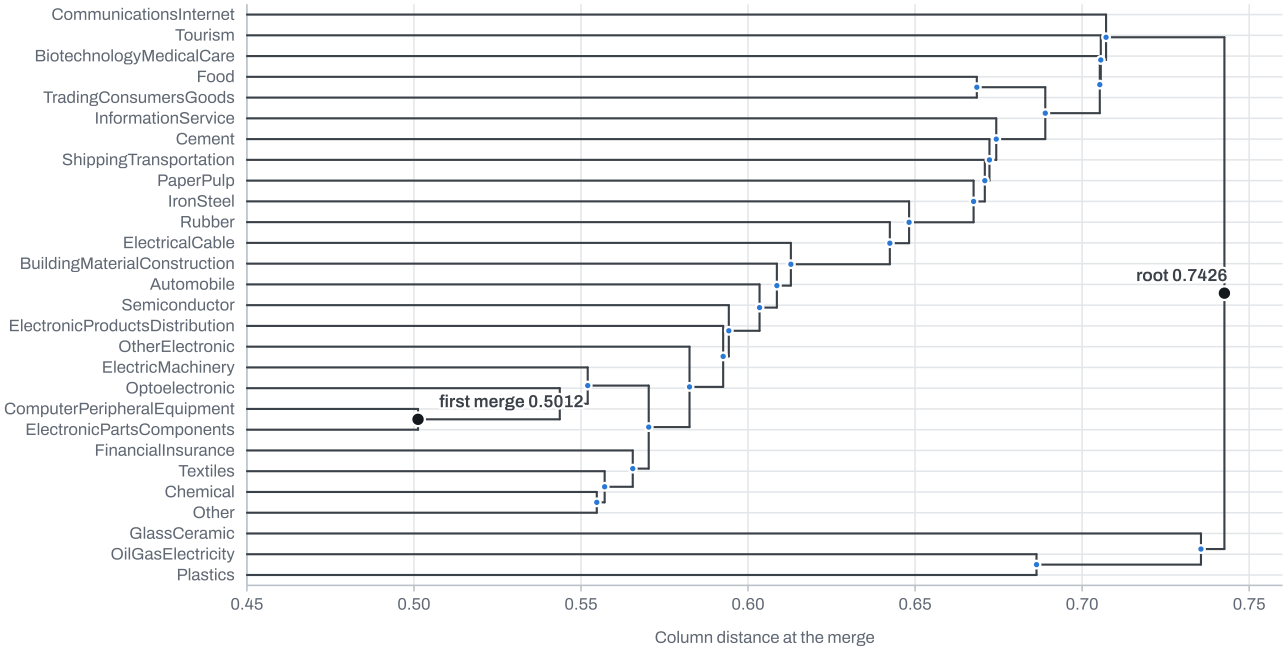


Note: daily price returns, 2 December 2008 to 8 October 2026, $T = 4{,}382$; price indices; descriptive. The order, top to bottom: CommunicationsInternet, Tourism, BiotechnologyMedicalCare, Food, TradingConsumersGoods, InformationService, Cement, ShippingTransportation, PaperPulp, IronSteel, Rubber, ElectricalCable, BuildingMaterialConstruction, Automobile.

Then: Semiconductor, ElectronicProductsDistribution, OtherElectronic, ElectricMachinery, Optoelectronic, ComputerPeripheralEquipment, ElectronicPartsComponents, FinancialInsurance, Textiles, Chemical, Other, GlassCeramic, OilGasElectricity, Plastics.

Source: FinMind `TaiwanStockPrice`, retrieved 11 October 2026; 524 closes in this sample (on 38 dates) replaced by their 13:30:00 values in `TaiwanStockEvery5SecondsIndex`, as in LM10. Computations: `data/lm31_numbers.py`.

Exhibit 4: U28's HRP tree



Note: single linkage on the full sample's column distance; sectors in Exhibit 3's order; in-sample. Source: as Exhibit 3.

PITFALL **Two codes, two trees**

The author's code passes the square matrix (d_{ij}) to SciPy's `linkage`, which reads its rows as observations and warns. So it builds the column-distance tree, at all 92 windows and on U28. Code that passes the condensed d_{ij} links on d itself, and the quasi-diagonal order differs at 17 of the 92 windows. The bisection cuts the list, not the tree (Example 6).

導讀 **先分群，再由上而下分配**

層次風險平價（hierarchical risk parity，HRP）不需要反矩陣（inverse matrix），也不需要最佳化（optimization）。第一步用相關係數（correlation coefficient）定義距離 d_{ij}，再算距離矩陣各行之間的歐氏距離（Euclidean distance），以單一連接叢集法（single linkage clustering）做層次叢集法（hierarchical clustering），畫成樹形圖（dendrogram）。第二步準對角化（quasi-diagonalization），讓相關高的資產排在一起。第三步遞迴二分（recursive bisection）：把排好的清單從中間切開，風險較低的一半分到較多權重。切的是清單而不是樹，所以樹上最先合併的一對也可能被切開。

KNOWLEDGE CHECK 3

Build the tree with LM8's $\sqrt{2(1-\rho_{ij})}$ in place of d_{ij}. What changes?

ANSWER

Every distance and column distance doubles. Single linkage only compares distances, so the merges stay and their heights double, on U28 and at all 92 windows.

5 R6 AND R7 OUT OF SAMPLE

- [ ] explain what risk-based weights assume, and why risk shares set in advance differ after the fact

At the 92 month-end signals R6 and R7 reset their weights (Exhibit 5). R6's move little, 00679B's between 29.7% and 37.4%. R7's move more, since a small change in the correlations can reorder its tree: 00679B's runs from 21.7% to 58.8%.

EXAMPLE 7

What R6's equal shares became

R6 sets every share to 20% on each trailing window. What shares did its weights carry afterwards?

SOLUTION

On the covariance of the 1,845 joint dates from 5 March 2019, the mean of R6's 92 target weights carries 20.5, 19.2, 19.2, 23.1 and 18.0% of the risk.

Each holding month's own target weights, on that holding month's covariance, give 00635U a share from 8.7% to 47.0% (10% and 90% quantiles) and 00679B one from −0.9% to 41.1%. Some share is negative in 29 of the 91 holding months. Equal risk is a statement about $\hat{\Sigma}$ before the fact.

Exhibit 5: R6 and R7 across the 92 signals

Fund	R6 weight, mean (min–max)	R7 weight, mean (min–max)	R7 share, mean	R6 share, ex post	R6 share by holding month, 10%–90%
0050	16.1 (13.3–18.3)	13.6 (6.6–21.8)	16.0	20.5	4.7 to 38.5
00646	18.4 (16.2–21.8)	20.0 (12.8–29.8)	21.9	19.2	2.1 to 30.6
00679B	32.8 (29.7–37.4)	33.7 (21.7–58.8)	26.6	19.2	−0.9 to 41.1
00635U	22.1 (18.6–27.5)	27.5 (13.7–34.8)	29.1	23.1	8.7 to 47.0
00642U	10.6 (8.5–12.9)	5.2 (2.8–8.3)	6.5	18.0	2.1 to 32.0
Realized volatility (SD)	10.32	10.14			

Note: %; out-of-sample, net of costs. R7's share is on each window's $\hat{\Sigma}$; its means, rounded separately, sum to 100.1. The ex-post share takes the mean of R6's 92 target weights on the covariance of the 1,845 joint dates from 5 March 2019. The last column takes each holding month's own target weights on that holding month's covariance. Realized volatility: the standard deviation (divisor T) of daily net returns from 5 March 2019, annualized with A_{w}. Source: as Exhibit 1.

Risk-based weights use $\hat{\Sigma}$ alone. They ignore the means and assume that the window's covariance still holds after the trade. Example 7 shows that it held roughly over the whole period but not within single holding months.

Over the 1,851 sessions from 5 March 2019, R6's daily net returns had a standard deviation of 10.32% a year, and R7's of 10.14%. LM30 reports 12.48% for R1 (this period's, not LM27's full-sample 11.49%), 9.96% for R3 and 9.97% for R5.

On U28 one split was fixed in advance: weights from the first 2,191 returns, held through the second (Exhibit 6). On the second half HRP, at 14.96%, is the lowest of 1/N and the three risk-based portfolios, and above both minimum-variance portfolios.

Exhibit 6: U28's halves: weights set on the first half

Portfolio	First half	Second half
1/N (LM30)	15.98	15.51
Inverse volatility	15.70	15.20
ERC	15.57	15.11
HRP	15.22	14.96
Long-only minimum variance (LM30)	12.74	12.14
Sample minimum variance (LM30)	11.40	11.78

Note: volatility, % a year, $A = 245.48$. The halves hold 2,191 returns each (LM11): 2 December 2008 to 6 October 2017, and 11 October 2017 to 8 October 2026. In-sample in the first column, out-of-sample in the second; price indices, covariance only. Source: as Exhibit 3.

López de Prado's own evidence is a Monte Carlo experiment: 10 series of 520 Gaussian observations, a 260-observation window, re-estimated and rebalanced every 22 observations, 10,000 runs. There HRP had a lower out-of-sample variance than the critical line algorithm, a quadratic optimizer (his abstract). Neither this experiment nor U28's split tests E5.

KNOWLEDGE CHECK 4

The mean of R6's 92 target weights put 23.1% of the realized risk in 00635U, not 20%. What does R6 assume about the covariance, and why can equal shares before the fact end unequal?

ANSWER

R6 uses only each trailing window's $\hat{\Sigma}$, so it assumes that this covariance still holds after the trade. On the later covariance the products $w_i(\Sigma \mathbf{w})_i$ no longer agree.

6 VOLATILITY TARGETING: R8

- ☐ compute a volatility-targeted exposure from an EWMA forecast, with its cap, cash and borrowing
- ☐ state the evidence on volatility-managed portfolios with its sample and design

R6 fixes the shares of risk but not its level. At signal t, R6's new weights on the window's 504 joint dates give a book with uncentred mean square $\bar{\sigma}_t^2$. Run through the same returns from $\bar{\sigma}_t^2$, LM26's EWMA (26.2), $\lambda = 0.94$, gives $\sigma_{t+1|t}^2$. R8 scales R6 by their ratio:

$$L_t = \min\left(1.5,\ \frac{\bar{\sigma}_t}{\sigma_{t+1|t}}\right), \qquad \mathbf{f}_t = L_t\,\mathbf{w}_t^{\mathrm{R6}}. \tag{31.7}$$

LM26 writes (26.2) for log returns; R8 applies it to daily simple returns in decimals, uncentred. Annual figures of $\bar{\sigma}_t$ and the forecast use the estimation frame's A. Cash, $1 - L_t > 0$, earns $r_f = 1.5\%$ a year, LM7's illustrative rate, and borrowing costs r_f plus 2 points, 3.5%, both accrued per session over A_{w}. Between trades the exposures drift, and (27.1)–(27.2) book returns, interest and costs. R8 keeps $1 + R_{p,t} > 0$ on all 1,851 sessions.

The financing rate, 3.5% a year, is the through case's illustrative rate, since no official one exists. LM34 gives the sources of the commission and the tax, and R8's growth at other borrowing rates. The tail risk of the scaled book, its value at risk and expected shortfall, is LM32's subject.

EXAMPLE 8

R8 at the first signal

On 27 February 2019, find R8's exposure.

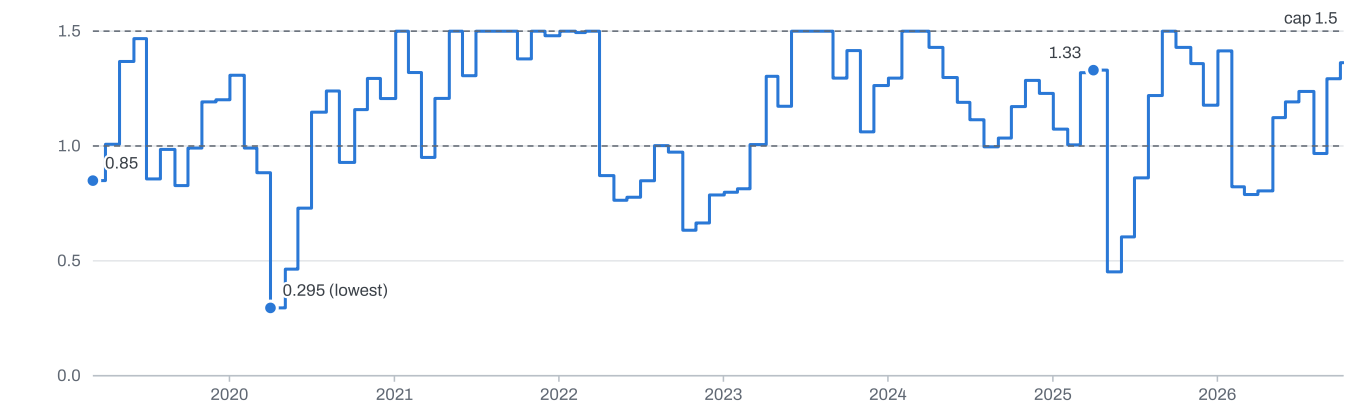
SOLUTION

The window gives $\bar{\sigma}_t = 0.0034991$ a day (5.45% a year) and the EWMA 0.0041204 (6.42%). So $L = 0.0034991/0.0041204 = 0.8492$, the exposures are 0.8492 times R6's weights, 0.1326, 0.1552, 0.2875, 0.2014 and 0.0725, and the cash, 0.151, earns 1.5%. After 504 days the start value's weight is $0.94^{504} = 2.9 \times 10^{-14}$.

Harvey et al. (2018) target a fixed 10% with a zero-mean EWMA of half-life 20 days, $\lambda = 0.5^{1/20} = 0.9659$; R8 targets $\bar{\sigma}_t$, with a half-life of 11.2 days (LM26).

Over the 92 trades R8's exposure averaged 1.14 (Exhibit 7): at the cap 14 times, above 1 at 62 and below 1 at 30. The uncapped ratio reached 2.14. Over the signals $\bar{\sigma}_t$ ran from 5.37% to 11.24% a year (mean 9.42%). The forecast ran from 3.66% (31 May 2019) to 31.48% (31 March 2020), which set the low of 0.295.

Exhibit 7: R8's exposure at its 92 trades



Note: L_t by (31.7) at each trade, drawn as a step. Between trades the held exposure drifts, to at most 1.53 on 22 August 2023, and is not plotted. Dashed at 1 and at the cap 1.5. Marked: 4 March 2019 (0.85), 1 April 2020 (0.295) and 1 April 2025 (1.33). Out-of-sample. Source: as Exhibit 1.

LM26's recursion extends to a covariance matrix, and for fixed weights its quadratic form is (26.2) on the book's returns:

$$\begin{aligned} \Sigma_{t+1|t} &= \lambda \Sigma_{t|t-1} + (1-\lambda)\mathbf{R}_t\mathbf{R}_t^\top, \\ \mathbf{w}^\top \Sigma_{t+1|t}\mathbf{w} &= \lambda\, \mathbf{w}^\top \Sigma_{t|t-1}\mathbf{w} \\ &\quad + (1-\lambda)(\mathbf{w}^\top \mathbf{R}_t)^2. \end{aligned} \tag{31.8}$$

So from a start with $\mathbf{w}^\top \Sigma_{\text{start}}\mathbf{w} = \bar{\sigma}_t^2$, R8's forecast equals $\mathbf{w}^\top \Sigma_{t+1|t}\mathbf{w}$; on E5 the two agree to rounding, below 10^{-14} of $\bar{\sigma}_t^2$. The matrix stays positive semidefinite (Problem 9). Multivariate GARCH is beyond this book.

The evidence. Moreira and Muir (2017) scale factor returns each month by a constant over the previous calendar month's realized variance. The constant equates the managed volatility with buy-and-hold's over the full sample, 1926–2015 for the market, size, value and momentum. Caps of 1 and 1.5 appear only as robustness checks. Sharpe ratios rise, because expected returns do not rise in proportion to volatility (their abstract).

Harvey et al. study more than 60 assets with daily data from as early as 1926. Their Sharpe ratio improves slightly for equity indexes and credit, hardly for bonds, currencies and commodities. The volatility of volatility falls and the left tail improves for almost all their futures, forwards and credit. For US 10-year Treasury futures the left tail is similar. Neither paper tests E5's funds.

> **PITFALL** **A target that moves with the book**
>
> R8 has no fixed target. It targets R6's own volatility over the last 504 joint dates, which ran from 5.37% to 11.24%. So R8 aims low after calm years and high after a crash.

導讀 **用預測波動率調整曝險**

波動率目標（volatility targeting）在預測波動率低時加碼、高時減碼。R8 的目標不是固定水準，而是 R6 自己過去 504 個共同交易日的波動率；預測來自指數加權移動平均模型（EWMA），兩者的比值就是曝險（exposure），上限 1.5 倍。曝險超過 1 就是槓桿（leverage），借款年息 3.5%；低於 1 的部分是現金，年息 1.5%。Moreira 與 Muir 用上個月的實現變異數（realized variance）縮放，常數由全樣本決定；Harvey 等人以固定 10% 為目標，研究 60 多種資產。這些研究都沒有檢驗 E5 的基金，結果不能直接套用。

KNOWLEDGE CHECK 5

(a) At the first signal $\bar{\sigma}_t = 5.45\%$ and the EWMA forecast is 6.42% a year. Find L_t and the cash. (b) Name two ways in which Moreira and Muir's managed portfolio differs from R8.

ANSWER

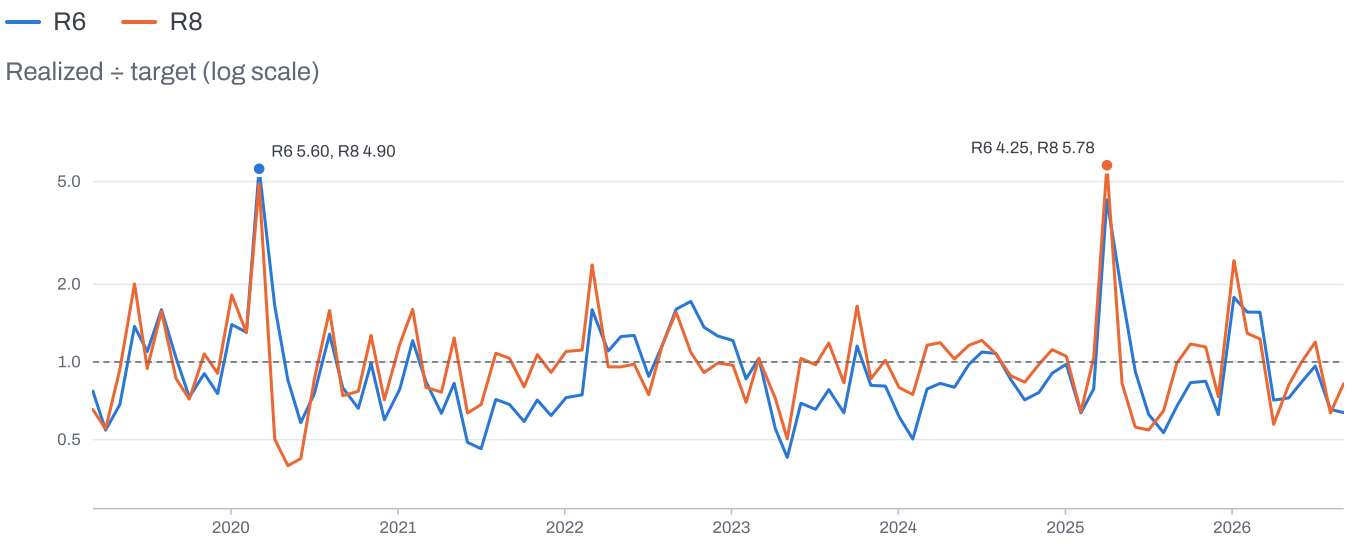
(a) The exposure is $5.45/6.42 = 0.849$, and the cash $1 - 0.849$, or 15.1%. (b) Any two. Moreira and Muir's portfolio scales by the inverse of the previous calendar month's realized variance, not a ratio of volatilities. Its constant uses the whole sample. Its main results have no cap.

7 JUDGING THE TARGET OUT OF SAMPLE

- [ ] judge a volatility target out of sample, and explain how periodic and risk-triggered rebalancing differ

A holding month runs from the session after a trade through the next trade, 12 to 23 sessions. The 91 complete holding months exclude the last 5 sessions. A holding month's realized volatility, the uncentred root mean square of its daily net returns, is divided by $\bar{\sigma}_t$ at its signal (Exhibit 8). What R8 lost on 7 April 2025 is LM35's replay; this module reports only its exposure and that holding month's volatility.

Exhibit 8: Realized over target volatility, 91 holding months



Note: log scale; dashed at 1. Each point, at a holding month's first session, is its realized RMS over $\bar{\sigma}_t$ at its signal. Marked: 3 March 2020 (R6 5.60, R8 4.90) and 2 April 2025 (R6 4.25, R8 5.78). Out-of-sample. Source: as Exhibit 1.

EXAMPLE 9

Ninety-one holding months

Compare R6's and R8's ratios, and score the window and the EWMA as forecasts of R6's next holding month.

SOLUTION

The median ratio is 0.83 for R6 and 0.98 for R8, but the SD of its logarithm (divisor 91), the spread across holding months, is 0.421 for both. Both have 11 holding months above 1.5; R6 has 21 below 2/3, R8 13. Over the whole period R8's daily net returns had a standard deviation of 11.52% a year, R6's 10.32%.

With V the holding month's mean squared daily return of R6, QLIKE (26.8) averages 0.691 for the window and 0.791 for the EWMA, with medians 0.165 and 0.087. The EWMA is lower in 56 holding months and higher in 35.

Two holding months decide the means. The EWMA's largest loss, by 13.3, is 27.50 (2 April → 2 May 2025, window 14.17), and its largest win, by 6.63, is 20.33 (3 March → 1 April 2020, window 26.96). Without the 2025 holding month the means are 0.541 (window) and 0.494 (EWMA); without both, 0.244 and 0.271. R8 moved the typical holding month to its target; it did not narrow the spread.

The 2025 holding month costs the EWMA so much for LM26's reason: EWMA has no long-run level, so a one-day forecast is also its forecast for every later day. A month-end rule then holds an old exposure through a price gap; GARCH's forecast (26.4) would revert (the lab tries it).

EXAMPLE 10

April 2025

Follow R8 from the signal of 31 March 2025 to its next trade.

SOLUTION

From $\bar{\sigma}_t = 7.84\%$ and the EWMA's 5.89%, $L_t = 1.33$. Continued daily on R6's 31 March weights held constant, the EWMA reads 6.34% on 2 April and 23.58% on 7 April (exposure 0.33). R8 trades next on 2 May, at 0.452. Its realized volatility, 45.3% (with the target's A), is 5.78 times the target, and R6's 4.25 times. A trigger could act only after 7 April's close, the first that showed the jump.

From 3 March 2020 R8 entered at 0.884, and its ratio, 4.90, was below R6's 5.60.

R8 re-estimates periodically, after month-ends only; a risk trigger also trades when the forecast moves far enough. The lab's trigger, fixed by the editor on 24 October 2026 before any lab result, updates the EWMA of R6's drifted book after each close. It trades at the next close when the implied exposure differs from the held one by more than 0.5.

The lab's trigger is an in-sample illustration. Choosing its threshold on these data would be a search of the kind that LM34 corrects.

R8's annual turnover and what it costs are LM34's; this module reports the size of its trades. Over the 91 trades after the first purchase R8 traded 9.2% one way on average, and the largest trade, 45.2%, was the cut of 2 May 2025.

On E5 the forecast that drives R8's leverage did not steady the risk, so the data do not support levering R6 to 1.5. And 91 holding months are a short record.

> **PITFALL** **A month-end rule cannot react to a gap**
>
> Between trades R8 keeps the exposure set at the last month-end: on 7 April 2025 it still carried the leverage of 1 April. A trigger reacts only after the day's loss.

導讀 **定期再平衡追不上跳空**

R8 只在每個月底訊號之後、下一個交易日收盤時再平衡（rebalancing），屬於定期再估計（periodic re-estimation）：兩次交易之間，曝險（exposure）只隨價格漂移（drift）。2025 年 4 月 7 日是清明連假後第一個交易日，R8 還帶著 4 月 1 日建立的槓桿（leverage），直到 5 月 2 日才調降。風險觸發（risk trigger）在預測曝險與持有曝險的差距超過門檻（threshold）時交易，但也只能在收盤之後反應，損失已經發生。91 個持有月份（holding month）裡，R8 讓典型月份的實現波動率（realized volatility）更接近目標，卻沒有縮小月份之間的離散程度。

KNOWLEDGE CHECK 6

(a) The EWMA had the lower QLIKE in 56 of 91 holding months, yet its mean QLIKE is the higher. How? (b) Could a risk trigger have cut R8's exposure before the loss of 7 April 2025?

ANSWER

(a) Its 56 wins gain 17.25 in all, 0.308 each; its 35 losses cost 26.37, 0.753 each. The net cost, 9.12 over 91 holding months, is the gap in means, 0.100. The holding month 2 April → 2 May 2025 alone costs 13.3, at 27.50 against 14.17: QLIKE punishes a forecast far too low. (b) No: a trigger acts only after a close, and 7 April's was the first to show the jump. A periodic rule waits for the month-end, a trigger for the next close after its threshold is crossed.

8 RISK-BASED ALLOCATION IN PYTHON

- ☐ compute equal-risk and risk-budget weights by Newton's method and coordinate descent, and check them with cvxpy
- ☐ build an HRP portfolio from a correlation matrix, and explain why implementations can give different trees

PYTHON

```python
import cvxpy as cp
import numpy as np
from scipy.cluster.hierarchy import leaves_list, linkage
from scipy.spatial.distance import squareform

def ccd(S, b, tol=1e-12):                # (31.4), S annualized
    y = 1 / np.sqrt(np.diag(S))
    while np.abs(y * (S @ y) - b).max() >= tol:
        for i in range(len(b)):
            r = S[i] @ y - S[i, i] * y[i]  # the other funds' part
            y[i] = (np.sqrt(r * r + 4 * S[i, i] * b[i]) - r) / (2 * S[i, i])
    return y / y.sum()

def newton(S, b):                        # as vol5_rules.erc
    y = 1 / np.sqrt(np.diag(S))
    y *= np.sqrt(1 / (len(b) * (y @ S @ y)))
    h = lambda v: 0.5 * v @ S @ v - b @ np.log(v)
    while np.abs(y * (S @ y) - b).max() >= 1e-13:
        g = S @ y - b / y
        d = np.linalg.solve(S + np.diag(b / y**2), -g)
        t, far = 1.0, np.abs(g * y).max() > 1e-6
        while (y + t * d).min() <= 0 or (
                far and h(y + t * d) > h(y) + 1e-4 * t * (g @ d)):
            t /= 2                        # halve until h falls enough
        y = y + t * d
    return y / y.sum()

# S: the annualized covariance, b: the budgets, x: the window's returns
y = cp.Variable(len(b))                  # the check: h in cvxpy
h = 0.5 * cp.quad_form(y, cp.psd_wrap(S)) - b @ cp.log(y)  # S is PSD
prob = cp.Problem(cp.Minimize(h))
prob.solve(solver=cp.CLARABEL, tol_gap_abs=1e-10, tol_gap_rel=1e-10,
           tol_feas=1e-10)
assert prob.status == "optimal" and y.value is not None

C = np.corrcoef(x, rowvar=False)         # x: T x N returns
C = (C + C.T) / 2                        # exactly symmetric
np.fill_diagonal(C, 1.0)
d = np.sqrt(np.clip((1 - C) / 2, 0, None))        # (31.5)
D2 = np.sqrt(((d[:, :, None] - d[:, None, :]) ** 2).sum(axis=0))
D2 = (D2 + D2.T) / 2
np.fill_diagonal(D2, 0.0)
Z = linkage(squareform(D2), method="single")      # checks pass
order = leaves_list(Z)                            # quasi-diagonal
# linkage(d, "single") reads the rows of the square d as observations,
# warns (ClusterWarning) and builds the column-distance tree as well
```

`squareform` requires exact symmetry and a zero diagonal, which `np.corrcoef` does not guarantee.

IN PYTHON **Companion lab**

The Polars-first notebook `LM31_lab.ipynb` rebuilds the risk shares, the four ERC solvers, HRP by hand and at every signal, U28's tree and halves, R8 and its holding months. It solves the computational practice problems and ends with a GARCH-scaled R8.

The notebook reads FinMind files that you download once with your own token. From the repository root, run `uv run python data/fetch_finmind.py`, then `uv run python data/make_lab_data.py`. FinMind's licence does not allow the book to redistribute the data. After that, the notebook runs offline.

KNOWLEDGE CHECK 7

Why would SciPy reject `squareform(d)` if `d` were computed from `np.corrcoef` without the code's two lines that clean `C`?

ANSWER

On some of E5's windows that `d` is asymmetric at machine precision (nonzero, below 10^{-15}), and on some its diagonal is not exactly 0. Either fault makes `squareform`'s checks reject it. The code symmetrizes `C` and sets its diagonal to 1 first, so its `d` passes.

SUMMARY

- Risk budgets (31.1) set $\mathrm{RC}_i = b_i\sigma_p$; ERC is inverse volatility (31.3) for two funds or one common correlation.
- For positive definite Σ and positive budgets the long-only risk-budget portfolio is unique, (31.2). Newton's method and CCD (31.4) find it, but a fixed point need not.
- HRP clusters on the column distance of d_{ij}, (31.5), and bisects the ordered list, (31.6).
- Shares set before the fact moved afterwards; some share was negative in 29 of 91 holding months.
- R8, (31.7), scales R6 to its window volatility, capped at 1.5.
- Targeting moved R8's median ratio to 0.98 but left the SD of the log ratio at 0.421; the published evidence uses other designs and samples.

PRACTICE PROBLEMS

Problems on data use E5's first window (504 joint dates to 27 February 2019) with divisor T, long-only weights summing to 1, shares RC_i/σ_p and annual volatilities.

1. Two funds have volatilities 20% and 10% and correlation 0.5 (constructed). The ERC weight of the first is *closest* to:

   A. 0.200

   B. 0.333

   C. 0.500

2. For the same two funds, the budget b_1 for which equal weights solve (31.1) is *closest* to:

   A. 0.714

   B. 0.667

   C. 0.800

3. Three funds have volatilities 10%, 20% and 40%, and every correlation is 0.3 (constructed). The ERC weight of the 40% fund is *closest* to:

   A. 0.048

   B. 0.333

   C. 0.143

4. Prove theorem (31.2), citing the Deep Dive for existence. Then explain why the fixed point of Example 3, with equal shares, does not contradict uniqueness.

5. Two uncorrelated funds have volatilities 20% and 10% and budgets 0.7 and 0.3 (constructed). (a) Show that $w_i \propto \sqrt{b_i}/\sigma_i$, give w_1 to four decimals and check the shares. (b) Show that CCD reaches it in one sweep.

6. Three assets have $\rho_{12} = 0.8$, $\rho_{13} = 0.1$, $\rho_{23} = 0.2$ and volatilities 20%, 25% and 10% (constructed). Build HRP by hand with Section 4's rules: lower-numbered member first, the first cluster number 4, a first half of $\lfloor n/2 \rfloor$ items. Give d_{ij}, the column distances, the order and the weights to four decimals; check with SciPy.

7. Single linkage on LM8's $\sqrt{2(1-\rho_{ij})}$ instead of d_{ij} gives:

   A. the same tree, with every height doubled

   B. a different tree

   C. the same tree and heights

8. A window gives $\bar{\sigma}_t = 8\%$ a year (constructed). (a) Find L_t for forecasts of 5% and 16%. (b) Next day the funds' book returns −3%, with cash at 1.5% and borrowing at 3.5% a year over $A_{\mathrm{w}} = 243.57$ sessions. Give each book's return in percent to four decimals; why does the cap bind?

9. Show that $\Sigma_{t+1|t}$ of (31.8) stays positive semidefinite from a positive semidefinite start, and that $\mathbf{w}^{\top}\Sigma_{t+1|t}\mathbf{w}$ is the EWMA of $(\mathbf{w}^{\top}\mathbf{R}_t)^2$ from a matching start. Why does the start matter little after

504 days?

10. The rules trade at the next close after 92 month-end signals from 27 February 2019, on windows of 504 joint dates, net of (27.2)'s commission and tax. The results are out-of-sample. Over 91 holding months R8's realized-to-target ratio has median 0.98 and SD of its logarithm 0.421, R6's 0.83 and 0.421. Which is supported?

    A. Targeting halved the spread across holding months, and R6's equal shares held.

    B. Targeting moved the typical holding month to its target and left the spread unchanged.

    C. The EWMA forecast beat the window in every holding month.

11. *(Python)* On the first window's annualized $\hat{\Sigma}$, compute R6 by CCD and by Newton with Example 3's starts and stopping rules. Check both with cvxpy (CLARABEL, tolerances 10^{-10}, status `optimal`, agreement 10^{-5}) and by their shares. Then run the fixed point as Example 3 does. Give weights to four decimals.

12. *(Python)* Compute R7's first weights with SciPy. Then link on d_{ij} directly, and count the signals where the quasi-diagonal order differs, over the 92 month-ends from 27 February 2019 to 30 September 2026.

SOLUTIONS

1. **B** is correct: by (31.3), $w_1 = (1/0.2)/(1/0.2 + 1/0.1) = 0.333$. A is inverse variance, $0.25/1.25$. C is equal weights.

2. **A** is correct: at 50/50, $\Sigma\mathbf{w} = (0.025, 0.010)$, so fund 1's share is $0.0125/0.0175 = 0.714$, the budget that 50/50 meets by (27.6). B uses stand-alone risk, $0.1/0.15$. C keeps only the variance terms, $0.01/0.0125$.

3. **C** is correct: with one correlation ERC is inverse volatility, (31.3), so $(1/0.4)/(10 + 5 + 2.5) = 0.143$. A is inverse variance, $0.0625/1.3125$. B is 1/N.

4. As in Section 3: h is strictly convex on $\mathbf{y} > \mathbf{0}$ by (29.2), so by (29.3)'s argument it has at most one minimizer. The Deep Dive shows that one exists. Its first-order condition $y_i(\Sigma\mathbf{y})_i = b_i$ gives (31.1) for $\mathbf{y}^*/\mathbf{1}^\top\mathbf{y}^*$, and a long-only solution rescales to a stationary point, the minimizer. The fixed point has negative weights, outside the set the theorem covers.

5. (a) With $\rho = 0$, (31.1) reads $w_i^2\sigma_i^2 = b_i\sigma_p^2$, so $w_i \propto \sqrt{b_i}/\sigma_i$ and $w_1 = 4.1833/(4.1833 + 5.4772) = 0.4330$. The variance terms $0.4330^2 \times 0.04 = 0.00750$ and $0.5670^2 \times 0.01 = 0.00321$ split 0.70/0.30. (b) With no covariance the sum in (31.4) is 0, so each update sets $y_i = \sqrt{b_i}/\sigma_i$, and one sweep reaches the solution.

6. By (31.5), d is 0.3162, 0.6708 and 0.6325; from the unrounded d the column distances are 0.4489, 1.0000 and 0.9622. Assets 1 and 2 merge at 0.4489 into cluster 4, and asset 3 joins at 0.9622, so the order is (3, 1, 2).

At the first cut asset 3's variance is 0.01, and the half [1, 2], weighted 25/41 and 16/41, has variance $73/1681 = 0.043427$, so asset 3 takes 0.81283. At the second cut asset 1 takes $0.0625/0.1025 = 0.60976$ of 0.18717. The weights are 0.1141, 0.0730 and 0.8128 (rounded separately; 0.9999 as printed).

7. **A** is correct: every distance doubles, and single linkage only compares distances, so the merges stay and the heights double. B and C each drop one fact.

8. (a) The ratio $8/5 = 1.6$ exceeds the cap, so $L_t = 1.5$; and $L_t = 8/16 = 0.5$. (b) The first book returns $1.5 \times (-3\%) - 0.5 \times 3.5\%/243.57 = -4.5072\%$, the second $0.5 \times (-3\%) + 0.5 \times 1.5\%/243.57 = -1.4969\%$. The cap binds because the uncapped ratio, 1.6, is above 1.5.

9. For any vector $\mathbf{a}$, the first line of (31.8) gives

$$\begin{aligned}\mathbf{a}^\top \Sigma_{t+1|t} \mathbf{a} = \lambda\, \mathbf{a}^\top \Sigma_{t|t-1} \mathbf{a} \\ + (1-\lambda)(\mathbf{a}^\top \mathbf{R}_t)^2,\end{aligned}$$

   and the right side is at least 0 when the previous form is. So induction from the start gives positive semidefiniteness, as in (16.4). With $\mathbf{a} = \mathbf{w}$ the line is (26.2) on $\mathbf{w}^\top \mathbf{R}_t$. After 504 days the start weighs $0.94^{504} = 2.9 \times 10^{-14}$.

10. **B** is correct: the median moved from 0.83 to 0.98 while the SD of the log ratio stayed at 0.421. A contradicts the equal SDs, and Example 7 shows that R6's shares did not stay equal. Nothing in the stem supports C, and Example 9 refutes it: the EWMA lost 35 of the 91 holding months.

11. The lab gives 0.1561, 0.1828, 0.3386, 0.2372 and 0.0854 (CCD in 12 sweeps, Newton in 6 steps), every share 20%. cvxpy reports `optimal` and agrees with Newton to below 10^{-6}. The fixed point from 1/N settles after 220 steps at −1.3069, −1.3060, 2.1321, 2.4082 and −0.9274, every share 20% (Example 3).

12. The lab gives 0.1131, 0.2189, 0.3443, 0.2852 and 0.0384 (Example 6); linking on d_{ij} changes the quasi-diagonal order at 17 of the 92 signals.

GLOSSARY

Term	中文	Meaning
Risk contribution	風險貢獻	$\mathrm{RC}_i = w_i(\Sigma\mathbf{w})_i/\sigma_p$; the contributions sum to σ_p (LM27).
Risk share	風險份額	RC_i/σ_p, a fund's share of the book's volatility on one covariance matrix.
Risk budget	風險預算	A target share $b_i > 0$ for each fund, summing to 1.
Risk-budget portfolio	風險預算投資組合	The long-only solution of (31.1).
Equal risk contributions (ERC)	等風險貢獻	The risk-budget portfolio with $b_i = 1/N$.
Risk parity	風險平價	Allocation by risk shares rather than by means.
Inverse-volatility weights	反波動率權重	$w_i \propto 1/\sigma_i$, (31.3).
Cyclical coordinate descent (CCD)	座標下降法	Minimizing h along one coordinate at a time, (31.4).
Fixed point	不動點	A point that an iteration maps to itself, such as weights $\mathbf{w}$ with $w_i \propto b_i/\beta_{i,p}$, the betas taken at $\mathbf{w}$.
Hierarchical clustering	層次叢集法	Merging clusters step by step into a tree.
Single linkage	單一連接叢集法	Cluster distance as the smallest distance between members.
Column distance	行向量距離	$\|\mathbf{d}_i - \mathbf{d}_j\|$, the Euclidean distance between columns i and j of (d_{ij}), (31.5).
Dendrogram	樹形圖	A drawing of the merges and their heights.
Quasi-diagonalization	準對角化	Ordering the assets along the tree.
Recursive bisection	遞迴二分	Halving the ordered list and splitting weight by (31.6).
Hierarchical risk parity (HRP)	層次風險平價	López de Prado's tree-based allocation.
Volatility targeting	波動率目標	Scaling a book by a target over a volatility forecast, (31.7).
Holding month	持有月份	The sessions from the day after a trade through the next trade.
Periodic re-estimation	定期再估計	Re-estimating and trading on a calendar.

Term	中文	Meaning
Risk trigger	風險觸發	Trading when a risk measure crosses a threshold.

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